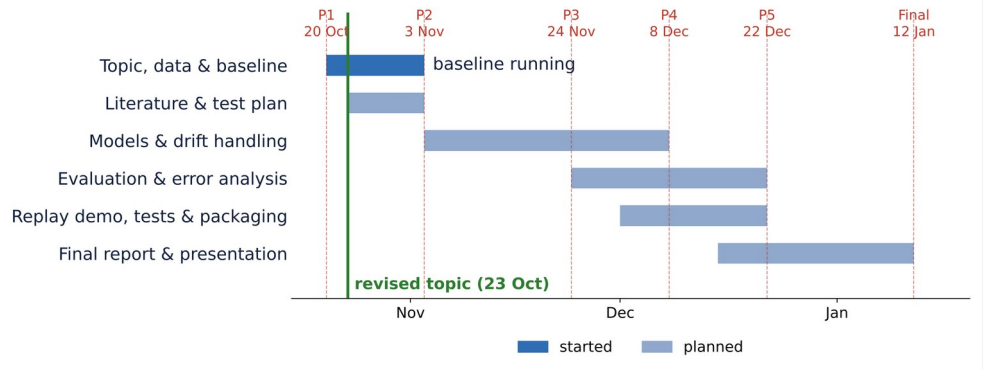
MDM2 – Case Study: Intelligent Systems in Production
One-Page Proposal

Team	Team 1
Members	Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas
Roles & Cases	Team lead and core code: Yu Han. Work packages are shared by all four members and tracked on the GitHub issues board.
Project Title	AI Decision Gate for False-Call Reduction in Automated Optical Inspection of Electronics Assembly
GitHub Repository URL	https://github.com/bodhi584/aoi-decision-gate
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Industrial Application (target domain/use-case)	Quality control in electronics assembly (SMT soldering): the step after automated optical inspection (AOI), where flagged boards go to manual inspection. A current industrial problem: on 21 Oct 2026 Siemens and NVIDIA announced an AI second check of AOI results at the Siemens electronics factory in Erlangen, targeting about 40 % fewer false calls and less manual verification. Based on the lecture example "Siemens – Quality Control and Defect Detection" (p. 18); real data from a Siemens production line in Bavaria.
Keywords (3–6)	quality control, automated optical inspection, false-call reduction, machine learning, neural networks, concept drift
Submission Date (YYYY-MM-DD)	2026-10-23
Gant Chart	
1) Problem Statement & Measurable Outcomes (3–4 sentences)	After soldering, an AOI camera system measures every solder joint against fixed limits, and flagged boards go to a manual inspection station. In the Siemens data about 98 % of flagged boards are false calls (good boards), which costs a large amount of manual work, and the line drifts over time. We build an AI decision gate: Input = AOI measurements of a flagged board; System = classifier trained on past manual-inspection results; Output = release the board, or send it to manual inspection. Targets, tested by replaying the later days in time order: at most 1 % of defective boards released, at least 25 % of false-call boards released (baseline: 8.2 %), and an alarm when performance drops.
2) Motivation & Industrial Relevance (2–3 sentences)	Every false call costs operator time and slows the line; every released defect reaches the customer. Siemens states that factories in high-wage countries need 7–9 % more productivity every year (Erlangen, Oct 2026). A reliable gate saves manual work without lowering quality.
3) Related Work Snapshot (2–3 key references)	Pfab et al. (2024): Siemens AOI dataset, 132 days, true defect vs false call; notes drift and label errors. Albayari et al. (2025): XGBoost, random forest and neural networks as a second classifier on Siemens AOI data. Schmitt et al. (2020): machine learning at Siemens Amberg predicts inspection results to reduce X-ray inspection. Our focus: evaluation in time order as on the real line, a fixed limit on missed defects, and drift handling.
4) Method & Feasibility (≤6 sentences)	Methodology: CRISP-DM (business understanding, data understanding, preparation, modelling, evaluation, deployment). Data: Pfab et al. (2024), Mendeley, CC BY-NC 3.0: 440,274 flagged solder joints on 39,742 boards, about 70 AOI measurements plus component type, error code and inspection type, labelled by manual inspection. Models: gradient-boosted trees and a neural network (PyTorch); threshold set so that at most 1 % of defective boards are released. Simulation: a day-by-day replay of the line (train on past days, decide on the next day, retrain on a schedule) measures released false calls, missed defects and saved inspections. Baseline already runs: trained on the first 70 % of days and catching 99 % of defective boards, it releases 8.2 % of false-call boards (29 % with a random split), so drift is the main challenge. Deliverables: a Python software prototype (command-line tool: train; decide per board → release or inspect; replay) with a replay web dashboard, tests on GitHub, demo video and report.
5) Milestones & Timeline (short table/list)	P1 23 Oct: revised topic, data set, baseline running P2 3 Nov: core code, baseline replay simulation, test plan, report P3 24 Nov: models and drift handling, unit tests and CI, issues board P4 8 Dec: full evaluation and error analysis P5 22 Dec: replay demo, demo video, run instructions, packaging Final 12 Jan: final report and presentation
6) Risks & Ethics (1–2 sentences)	Conditions and limits: one Siemens line, five inspection types; measurements are anonymised (scaled 0–1, no physical names), labels may contain human errors, and there are no images. Data licence is non-commercial (course use only); no personal data. Main risk: drift limits the release rate (fallback: report which drift handling works and why).